

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Total Marks: 40

Question Count: 5

Duration :60 Minutes

Code of Conduct

I BRIJESH.T/(192521391) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:  _____

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.
5. **Agriculture Smart Farming Problem**
A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time

field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

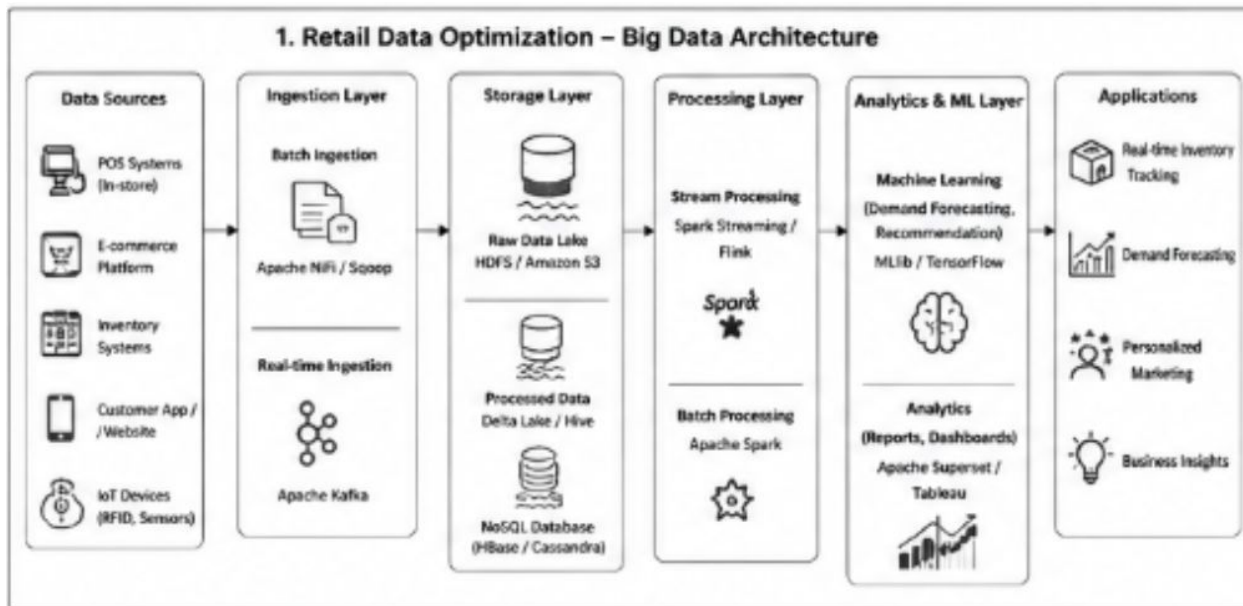
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

ANSWERS:

1. Retail Data Optimization Problem

A retail chain can use a big data architecture to collect and analyze sales, inventory, and customer behavior data from physical stores, point-of-sale systems, websites, and mobile applications. Apache Kafka or AWS Kinesis can be used for real-time data ingestion. Apache Flink or Spark Streaming can process the incoming data and provide real-time inventory updates.



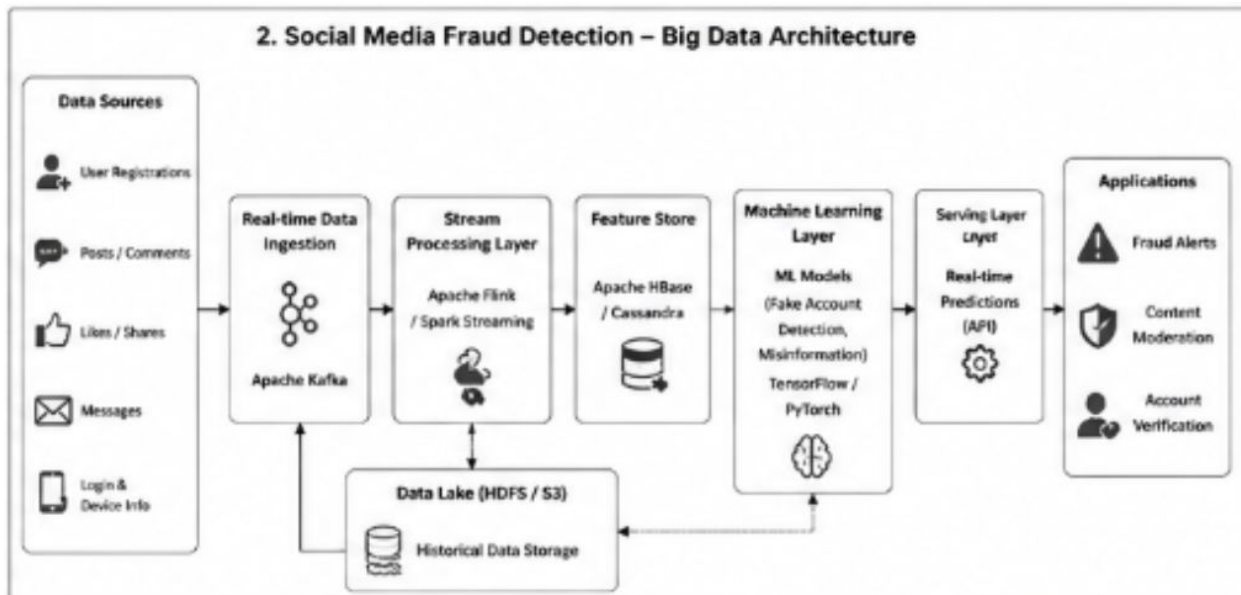
The processed data can be stored in a cloud data lake such as Amazon S3 or Azure Data Lake. Apache Spark can be used for large-scale batch processing and historical analysis. A data warehouse such as Snowflake, BigQuery, or Amazon Redshift can be used for business intelligence and reporting.

For demand forecasting, models such as ARIMA, Prophet, XGBoost, or LSTM can be used. These models can consider historical sales, seasonal trends, holidays, promotions, prices, and other factors. Customer segmentation can be performed using clustering algorithms such as K-means.

For personalized marketing, recommendation algorithms such as collaborative filtering, content-based filtering, and hybrid recommendation systems can be used. The final system helps the company maintain optimal inventory, reduce stockouts, forecast demand accurately, and provide personalized offers to customers.

2. Social Media Fraud Detection Problem

A social media platform can build a scalable big data system to process posts, comments, likes, shares, and user activities in real time. Apache Kafka or Apache Pulsar can be used to collect billions of user events and transmit them to the processing system.



Apache Flink or Spark Structured Streaming can process these events continuously. The system can calculate features such as posting frequency, account age, number of followers, IP address patterns, device information, and unusual activity patterns.

Machine-learning algorithms can then analyze these features to identify fake accounts. Algorithms such as Random Forest, XGBoost, neural networks, and graph neural networks can be used. Graph analytics is especially useful because fake accounts often form groups that interact with each other in coordinated ways.

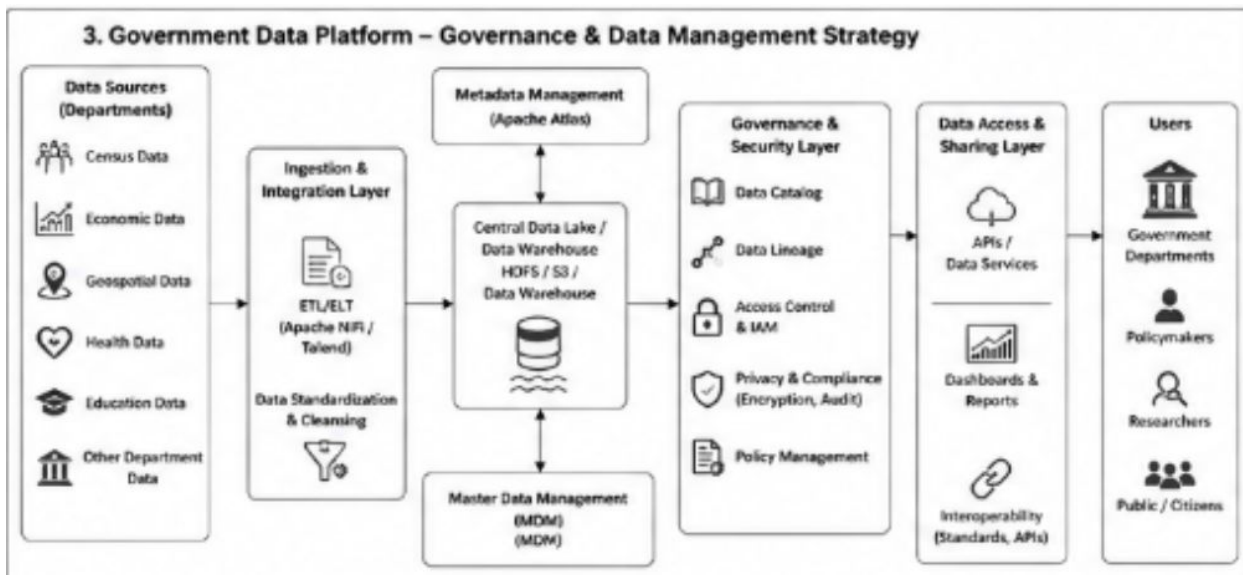
For misinformation detection, Natural Language Processing models can analyze posts and comments. Transformer-based models can classify suspicious content and identify potentially misleading claims. Image and video analysis can also be incorporated when required.

The system can generate a risk score for each account or piece of content. Low-risk content can be allowed, suspicious content can be sent for manual review, and high-risk accounts or content can be

restricted or blocked. This combination of stream processing and machine learning provides fast and scalable fraud detection.

3. Government Data Platform Problem

A government data platform should integrate census, economic, geospatial, and departmental datasets into a common data environment. Data can be collected using APIs, ETL pipelines, and other data-integration technologies and stored in a secure data lake or lakehouse.



Data standardization is important because different government departments may use different formats and definitions. Common data standards, identifiers, metadata, and data dictionaries should therefore be established. Master Data Management can be used to maintain consistent information about important entities such as locations, organizations, and administrative regions.

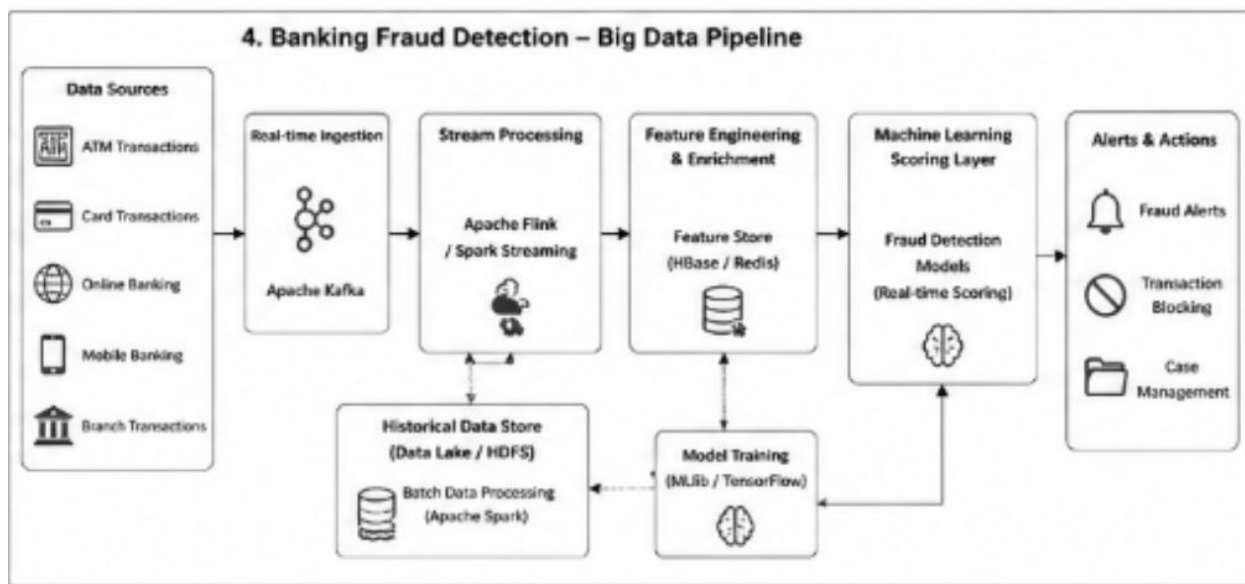
A central data catalog should provide information about available datasets, their owners, definitions, quality, update frequency, and access restrictions. Data-quality processes should identify missing values, duplicates, incorrect formats, and inconsistent information.

Security and privacy are critical because government datasets may contain sensitive information. Encryption, role-based access control, multi-factor authentication, anonymization, data masking, and audit logging should be implemented. Access should be provided according to the user's role and purpose.

A data governance framework should define data ownership, stewardship, privacy, security, retention, sharing, and quality policies. APIs and standardized data-exchange mechanisms can allow different government departments to securely access shared information. This approach provides interoperability while maintaining privacy, security, and data accountability.

4. Banking Fraud Detection Problem

A bank can use a real-time big data architecture to analyze millions of transactions from ATMs, point-of-sale terminals, mobile banking, and internet banking. Apache Kafka or Apache Pulsar can be used to ingest transaction events at high speed.



Apache Flink can process transactions in real time with low latency. During processing, features such as transaction amount, location, time, merchant, device, transaction frequency, and customer history can be generated.

Machine-learning models such as Logistic Regression, Random Forest, XGBoost, and neural networks can be trained using historical transaction data. These models can identify patterns associated with previously known fraudulent transactions. Anomaly-detection algorithms such as Isolation Forest and Autoencoders can identify unusual transactions that may represent new types of fraud.

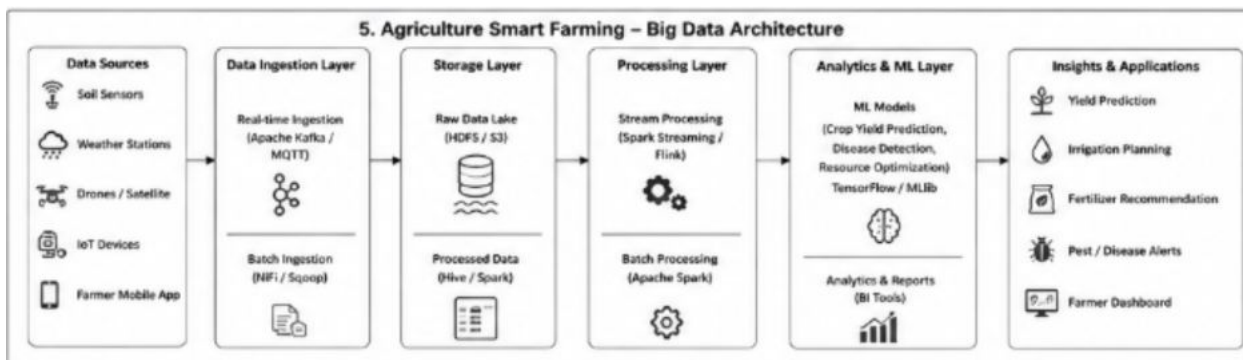
A rule engine can also be used for known fraud patterns. For example, multiple high-value transactions from geographically distant locations within a short period may be considered suspicious. Graph analytics can identify relationships between suspicious accounts, cards, devices, and merchants.

Historical transaction data can be stored in a data lake and processed using Apache Spark for model training and long-term analysis. The real-time system generates a fraud risk score for every transaction. Depending on the score, the transaction can be approved, sent for additional verification, or blocked.

This architecture provides high throughput, low latency, and accurate fraud detection while continuously improving the models using new transaction data.

5. Agriculture Smart Farming Problem

A precision-agriculture system can collect data from soil sensors, weather stations, drones, satellites, farm machinery, and crop-management systems. IoT technologies such as MQTT can be used to transmit sensor information, while Apache Kafka can be used to handle large volumes of streaming data.



Apache Flink can process real-time sensor data and generate immediate alerts. Apache Spark can process historical agricultural data for large-scale analysis. The collected data can be stored in a cloud data lake such as Amazon S3 or another distributed storage system.

Machine-learning algorithms can be used to predict crop yield based on soil conditions, weather, crop type, historical yield, and satellite information. Irrigation models can determine how much water is required based on soil moisture, weather conditions, and crop growth stages.

Computer-vision models can analyze drone and satellite images to detect crop diseases, pests, and areas of crop stress. GIS technology can be used to divide farms into different zones according to soil quality, moisture, nutrient levels, and crop health.

The system can provide farmers with recommendations about irrigation, fertilizer application, pest control, and harvesting. Real-time alerts can also be generated when abnormal conditions are detected. This helps farmers increase crop yield, reduce water and fertilizer usage, lower costs, and improve overall resource management.